

Bayesian :

Una *prior* (distribución a priori) en estadística bayesiana es la distribución de probabilidad que representa el conocimiento o creencia previa sobre un parámetro desconocido **antes** de observar los datos actuales.

Structure learning

Structure learning refers to the computational process of identifying the underlying, hidden architecture or relationships (structure) within data, rather than just optimizing parameters for a predefined model.

Structure learning is the procedure to determine causal relationship directions from observational data only and representing these as a (causal) graph. The basic idea emerged from Wright (1921) as path analysis.

ANOVA (Analysis of Variance)

- **Purpose:** Tests whether the means of three or more groups differ significantly.
- **Components:** Uses at least one categorical independent variable and one continuous dependent variable.
- **Example:** Comparing the test scores (dependent variable) of three different teaching methods (independent variable).
- **Mechanism:** Partitions the total sum of squares into variance due to main effects and variance due to error.
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ANCOVA (Analysis of Covariance)

- **Purpose:** Compares group means while controlling for the influence of an uncontrolled continuous variable (the covariate).
- **Components:** Includes a categorical independent variable, a continuous dependent variable, and one or more covariates (e.g., age, pre-test scores).
- **Example:** Comparing the test scores (dependent variable) of three teaching methods (independent variable) while controlling for students' prior knowledge (covariate).
- **Benefit:** Reduces error variance, increasing the statistical power of the test.

Key Differences Between ANOVA and ANCOVA

Feature	ANOVA	ANCOVA
Variables	Categorical IV, Continuous DV	Categorical IV, Continuous DV + Covariate(s)
Control	No control for external factors	Controls for covariates (e.g., baseline)
Goal	Simple group comparison	Comparison adjusted for external variance
Power	Lower if uncontrolled variables exist	Higher by reducing error term

Based on standard notation in probability and statistics, $p_{C,E}(c, e)$ represents the **joint probability mass function (PMF)** or **joint probability density function (PDF)** of two random variables, C and E . 

Here is a breakdown of the notation:

- p : Represents the probability or probability density function.
- C, E (Subscript): Specifies that this function applies to the random variables C and E together (the *joint* behavior).
- (c, e) (Parentheses): Indicates the specific values (c and e) at which the function is evaluated.

Meaning

This function calculates the probability that:

1. The random variable C takes on the specific value c , **AND**
2. The random variable E takes on the specific value e .

For discrete variables, $p_{C,E}(c, e) = P(C = c \text{ and } E = e)$.

Bayesian Graph :

For the graph:

Weather → Traffic → Late



We get:

$$P(\text{Weather, Traffic, Late}) = P(\text{Weather}) \cdot P(\text{Traffic} \mid \text{Weather}) \cdot P(\text{Late} \mid \text{Traffic})$$

A Gaussian Process prior =

"A distribution over possible functions, where any finite set of points follows a multivariate normal distribution"

A Gaussian process is a collection (of an infinite number) of Gaussian random variables that have some joint multivariate gaussian distribution $p(x_1, x_2, x_3, \dots)$. There is literally nothing else to them, every single fact or technique involving them follows from this.

The only thing that separates a gaussian process from a multivariate gaussian distribution is that the random variables in a gaussian process are indexed according to something like time or space. For example you might have a gaussian process written as $X(t)$; this just means that each value of 't' indexes a distinct gaussian random variable $X(t)$.

This is all easier to understand by thinking in terms of discrete sets of random variables. A gaussian process is just the continuous limit of this.

The arrow in " $X \rightarrow Y$ " signifies X has an effect on Y, or in other words, X is a cause of Y.

A **Directed Acyclic Graph (DAG)** is:

A graph with **directed edges (arrows)** and **no cycles**

The core problem

$$\text{Individual effect} = Y(1) - Y(0)$$

We only ever observe one of these. The other is the counterfactual — the world that didn't happen. This is what everything else is trying to reconstruct.

ATE

$$E[Y(1) - Y(0)]$$

Average effect across everyone. "What would happen if we ran this campaign for all users?"

ATT

$$E[Y(1) - Y(0) \mid T=1]$$

Effect only among those who were actually treated. "Did the campaign help the users who saw it?"

ITT

$$E[Y \mid \text{assigned}=1] - E[Y \mid \text{assigned}=0]$$

Effect of being assigned to treatment, regardless of whether people actually

complied. The standard estimand in most business experiments.

Why naive comparisons fail — selection bias

$$E[Y|T=1] - E[Y|T=0] = ATE + \text{Selection bias}$$

The raw difference in outcomes between exposed and unexposed conflates the true causal effect with pre-existing differences between the groups.

Where selection bias comes from

People who were exposed to the campaign are systematically different from those who weren't — they are younger, more active, more likely to subscribe anyway. The algorithm chose them for a reason. Any comparison that ignores this is mixing the causal effect of the campaign with the pre-existing advantages of the people it targeted.

The key assumption for observational methods

$$\{ Y(0), Y(1) \} \perp T \mid X$$

Conditional independence — once you control for observed characteristics X , treatment assignment is as good as random. Also called unconfoundedness or selection on observables.

METHOD

CORE IDEA

ASSUMPTION

BREAKS WHEN

Regression adjustment

Add confounders as control variables. The coefficient on treatment captures variation unexplained by those controls.

Linear relationship between confounders and outcome. All confounders measured.

Unobserved confounder exists. Groups overlap so poorly that regression is extrapolating.

Matching

For each treated unit, find the most similar untreated unit. Compare outcomes within matched pairs. Discard unmatched units.

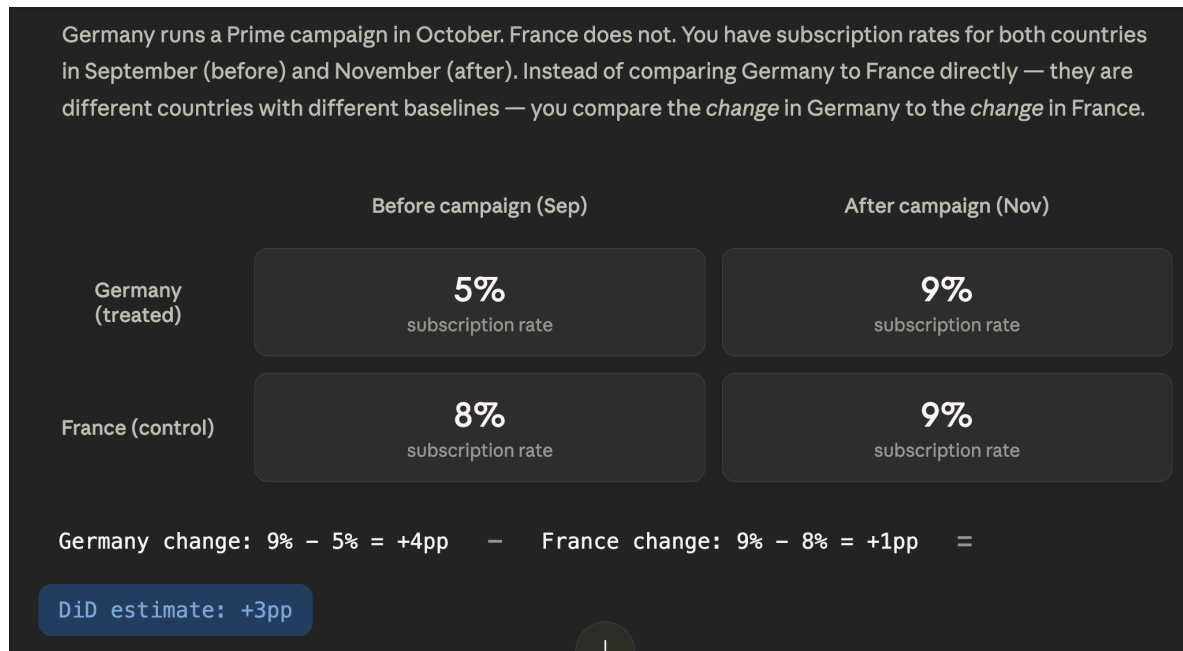
All confounders measured. Enough untreated units exist to find close matches.

Too many confounders (curse of dimensionality). Too few untreated units to match against.

The curse of dimensionality refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces (often hundreds of dimensions) that do not occur in low-dimensional settings. As dimensions increase, data becomes sparse, making it harder to find patterns, causing distances between points to become indistinguishable, and leading to model overfitting

Difference-in-Differences (DiD) — the core idea

The problem with regression and matching was: people who were exposed to the campaign were fundamentally different people from those who weren't. We couldn't find a fair comparison.



Synthetic control solves this by building a custom "synthetic Germany" — a weighted blend of other countries that, together, match Germany's pre-campaign trajectory as closely as possible.

No single country perfectly mirrors Germany. France has higher baseline rates. Spain has different seasonality. Italy has a different growth trend. But a carefully chosen blend — 40% France + 35% Netherlands + 25% Italy — might track Germany almost perfectly in the pre-campaign period. Once that blend is built, you use it as the counterfactual for what Germany would have looked like without the campaign.

Layer 2 — The parallel trends assumption in depth

$$E[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid \text{Treated}] = E[Y(0)_{\text{post}} - Y(0)_{\text{pre}} \mid \text{Control}]$$

In the absence of treatment, the treated group would have followed the same trend as the control group

The DiD regression equation

$$Y_{it} = \alpha + \beta \cdot \text{Treated}_i + \gamma \cdot \text{Post}_t + \delta \cdot (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

Y_{it}

Outcome for unit i at time t — e.g. subscription rate for country i in week t

α

Baseline level — the control group's outcome in the pre period

$\beta \cdot \text{Treated}_i$

How much the treated group differs from control on average, before treatment.

Captures pre-existing level differences. $\beta = 1$ if Germany, 0 if France.

$\gamma \cdot \text{Post}_t$

How much both groups change from pre to post — the shared time trend. $\gamma = 1$ after campaign launch, 0 before.

$\delta \cdot (\text{Treated} \times \text{Post})$

This is the DiD estimator. The coefficient on the interaction term. It captures the extra change in the treated group, above and beyond the shared time trend.

δ is your causal estimate.